

Yucheng (Cecilia) Zhang

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EDUCATION

Northwestern University

M.S. in Computer Science

Aug. 2026 – May 2028

Evanston, IL

The University of Texas at Austin

B.S. in Mathematics, Minor in Computer Science | GPA: 3.74/4.00

Aug. 2023 – May 2026

Austin, TX

Relevant Courses: Data Structures, Stochastic Processes, Elements of Software Design, Data Analytics, Database Systems, Discrete Math, Real Analysis, Probability and Statistics, Artificial Intelligence, Elements of Networking

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, R, MATLAB, SAS

Frameworks & Libraries: React, Flask, Node.js, REST APIs, Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, Matplotlib

Databases & Cloud: PostgreSQL, SQL Server, Oracle, Azure Synapse, Databricks, Delta Lake, AWS, Docker, Power BI

AI/ML & Tools: LLM APIs, OpenAI, Gemini, Claude, LangChain, CrewAI, MNE, Git, SQLAlchemy, Alembic, Pytest, Jest

WORK EXPERIENCE

Enterprise Risk Analyst Intern

Toyota Industries Commercial Finance, Inc. | Python, SQL Server, Azure, Power BI

May 2025 – Dec. 2025

Dallas, TX

- Developed an end-to-end credit risk analytics pipeline integrating PayNet inquiries and internal loan servicing data through SQL Server ETL, borrower and asset feature engineering, Python modeling, and monthly reporting automation.
- Validated models with ROC/AUC, confusion matrices, and feature importance, achieving **AUC > 0.82** across Class 8 trucks, trailers, and used-vehicle loans and improving high-risk case identification by approximately **3–4%**.
- Conducted backtesting by comparing predicted risk buckets against observed outcomes, revealing up to **3.7× higher default rates** in high-risk groups and informing credit scorecard segmentation refinements.
- Automated recurring model refreshes and Power BI reporting workflows, reducing manual reporting effort by 10% while supporting portfolio risk monitoring across truck, trailer, and used-vehicle segments.

Financial Data Development Engineer – Fraud Detection

Caitong Securities Ltd | SQL, Oracle, Stored Procedures

Jul. 2024 – Aug. 2024

Hangzhou, China

- Designed and optimized SQL-driven ETL pipelines processing **10M+ daily transactions** with window functions, subqueries, and stored procedures, improving extraction efficiency by 5%.
- Implemented automated fraud-detection logic with parameterized thresholds and modular stored procedures, reducing false positives by 2% for transaction-monitoring and AML-related workflows.
- Developed a customer risk scoring engine by integrating multiple transactional and behavioral data streams, enhancing fraud detection precision and financial data monitoring.

Software Engineering Intern – Customer Analytics

Joblogic-X Corporation | Python, APIs, Git

Nov. 2023 – Jun. 2024

Dallas, TX (Hybrid)

- Engineered a scalable data pipeline to scrape and process XHS social-media data and integrate it with transactional order systems for customer analytics.
- Implemented modular ranking and clustering algorithms in Python, generating reproducible customer-priority lists for marketing and engagement workflows.
- Integrated a KOC scoring module into Meet Fresh's customer-facing chatbot through REST APIs and maintained Git-based deployment workflows for reproducible updates.

Data Scientist Assistant Intern – Image Restoration

Software Engineering Center, Chinese Academy of Sciences | TensorFlow, Cloud, GFPGAN

Jul. 2023 – Sep. 2023

Beijing, China

- Deployed image restoration modules on Alibaba Cloud via SSH, implementing VAE loss, Gaussian filters, and LSGAN functions for scalable reconstruction workflows.
- Constructed a TensorFlow 2 CNN, partitioned 3.4k images into train/validation/test sets, and achieved **80% accuracy** across 5 image categories.
- Integrated GFPGAN APIs for automated face restoration and color correction, reducing processing costs by 5% and supporting model-comparison reporting.

RESEARCH EXPERIENCE AND PROJECTS

Echo – Interactive AI Voice Simulation System

University of Texas at Austin | Mentor: Prof. Bo Xie | Next.js, TypeScript, OpenAI API

Jun. 2026 – Present

Austin, TX

- Built an event-driven voice simulation system that generates context-aware responses from user pauses, errors, skipped fields, and task switching using session state, trigger rules, browser TTS, and OpenAI APIs.
- Designed cooldown, interval, and event-priority logic to prevent repetitive playback and make voice feedback responsive to task state, interaction timing, and user behavior.

CodeAssist – AI Feedback System GitHub	Aug. 2025 – Present
<i>University of Texas at Austin Mentor: Prof. Kia Teymourian React, Flask, PostgreSQL, Docker</i>	Austin, TX
• Built a full-stack AI feedback system for programming courses, connecting student code editing, file uploads, submissions, and chat-based AI feedback for debugging, complexity reasoning, implementation ideas, and test-case correctness. • Developed Flask REST APIs and PostgreSQL models for AI feedback settings, instructor controls, request limits, cooldown policies, role-based access, JSON serialization, and Alembic-backed schema changes. • Implemented LLM context filtering so assignment descriptions, student code, uploaded files, test cases, execution results, and student output are shared only when allowed by instructor-defined policies. • Standardized provider workflows across OpenAI, Gemini, Claude, and local models, including model selection, API configuration, timeout handling, and assignment-level prompt behavior.	
Researcher & TA – Biological Data Analytics	Jan. 2024 – Dec. 2024
<i>University of Texas at Austin Mentor: Prof. Becca Young R, Linux, AWS, Docker</i>	Austin, TX
• Processed and analyzed 250GB of RNA-seq data using Linux, Docker, and AWS workflows, improving processing speed by 6% and reducing storage costs by 5%. • Applied Salmon, RSEM, Trinity, and samtools for transcript quantification, achieving a 98.7% mapping rate and ensuring reproducibility across datasets. • Performed Voom/Limma differential-expression analysis, identified 48 key genes across time-series samples, and automated normalization, PCA, and volcano-plot workflows in R/ggplot2. • Collaborated as a teaching assistant to train incoming FRI students on reproducible RNA-seq pipelines and data literacy.	
FinChat – AI-Powered Financial Analysis System	Jan. 2023 – May 2024
<i>Southern Methodist University Prof. Labiba Jahan Python, FinGPT, LangChain</i>	Dallas, TX (Remote Hybrid)
• Built LLM workflows for financial report analysis by extracting and processing 100+ SEC filings from EDGAR into a structured data lake for metric extraction, trend detection, and company-level comparison. • Implemented LangChain/CrewAI workflows for contextual financial interpretation, improving filing-analysis accuracy by 3% and reducing manual review time for filing analysis. • Collaborated with a cross-university research team to connect FinGPT-based retrieval, filing preprocessing, and financial-question workflows into a reusable analysis prototype.	
Kiwifruit Quality Grading Project – Machine Learning Member	Aug. 2022 – May 2024
<i>Huazhong Agricultural University Mentor: Prof. Yan Tong Python, Computer Vision</i>	Wuhan, China (Hybrid)
• Contributed to an automated fruit grading pipeline using color, texture, shape, and feature-fusion methods; benchmarked model variants and analyzed results for the ML methodology section. • Compared feature combinations to improve grading accuracy and interpretation for research later published in <i>Expert Systems with Applications</i>.	
Machine Learning Research Assistant – Neuroengineering Lab	Jan. 2023 – May 2024
<i>Remote Mentor: Dr. Kunqiang Qing Python, MATLAB, MNE</i>	Remote
• Processed 35GB of multimodal EEG/fNIRS data for signal cleaning, visualization, feature extraction, connectivity analysis, and statistical testing. • Applied statistical tests and feature-extraction methods to evaluate cognitive impairment patterns and emotion-recognition signals across experimental conditions. • Evaluated RF, SVM, KNN, and XGBoost models, contributing to peer-reviewed work in <i>Frontiers in Neurology</i>, <i>Frontiers in Neuroscience</i>, and <i>IEEE ICIPCA</i>.	

LEADERSHIP AND COMMUNITY

Leader – Austin Chinese Campus Christian Fellowship	Aug. 2023 – Jun. 2026
<i>University of Texas at Austin</i>	Austin, TX
• Organized weekly Bible studies, retreats, hiking trips, and cultural celebrations; coordinated transportation and scheduling to improve group communication and inclusion. • Facilitated small-group discussions and welcomed new members through consistent event planning, follow-up communication, and peer support for students joining the group.	
Volunteer – Art From the Streets	Jan. 2026 – Present
<i>Community art programming for artists experiencing homelessness or housing insecurity</i>	Austin, TX
• Supported event preparation, artwork organization, and visitor engagement for community exhibitions featuring artists experiencing homelessness or housing insecurity. • Assisted volunteers and visitors during public art events, helping maintain a respectful and welcoming environment for participating artists and community guests.	

PUBLICATIONS

- Miao, H., Wang, Z., **Zhang, Y.**, ... & Tong, Y. (2025). New kiwifruit quality grading methods based on image multi-feature fusion and ResTNet model. *Expert Systems with Applications*, 129507.
- Xu, M., **Zhang, Y.**, Zhang, Y., Liu, X., & Qing, K. (2024). EEG biomarkers analysis in different cognitive impairment after stroke. *Frontiers in Neurology*, 15, 1358167.
- Zhang, Y.**, Zhang, Y., Jiang, Z., Xu, M., & Qing, K. (2023). The effect of EEG and fNIRS in digital therapy of Alzheimer’s disease. *Frontiers in Neuroscience*, 17, 1269359.
- Zhang, Y.** (2023). EEG-Based on Emotion Recognition Using Machine Learning. *IEEE ICIPCA*, 226–232.
- Li, D., Zhang, Z., & **Zhang, Y.** (2023). Public compassion under the mediation of suffering: Taking the videos of “Second Uncle” on social media as an example. *Media and Communication Research*, 4(5), 46–58.